

Activity 3: Selecting an appropriate Object Oriented Methodology.

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i) compare two object-oriented methodologies
suitable for this project.

Booch Method

- * Focuses on object oriented design and implementation.

- * Suitable for large and complex software projects.

- * Use detailed diagrams to represent objects and classes.

UML Based Development

- * Uses standard UML diagrams such as Use Case, class, sequence and activity diagrams.

- * Easy to understand and widely accepted.

- * Supports iterative and incremental development.

- * Easier to modify.

2) Recommended the best methodology and justify your answer.

Recommended Methodology : UML Based development
Justification :

- * The client frequently changes requirements
- * Support updation and iterative development making changes easier.

- * Therefore UML based development is the best choice for a Smart Parking management System.

3) Explain how object modelling helps during system analysis.

Object modelling helps by:

- * Identifying the main objects in the system.

- * Defining the attributes and methods of each object.

- * Showing relationships between objects and improving understanding of system requirements.

4) Identify possible classes and their relationships.

Classes:

- * User
- * Vehicle
- * Parking Spot
- * Parking Ticket
- * Payment
- * Parking Manager

Relationships:

- * A user owns one or more vehicles.
- * A vehicle is assigned a parking slot.
- * A parking ticket is used to make the required payment.
- * The Parking Manager manages parking slots and parking tickets.

5) Discuss how the selected methodology supports requirement changes.

UML Based Development supports requirement changes by:

- * Allowing diagrams to be updated without redesigning the entire system
- * Supporting iterative development and continuous improvements.
- * Making it easy to add and remove features before broking.
- * Reducing development time and maintainance effort.